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Review

How Can Grazing Mitigate Wildfires? A Review of Fuel Management, Ecological Trade-Offs, and Adaptive Frameworks

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Abstract

Under the influence of multiple factors such as climate change and human activities, the frequency, intensity, and destructiveness of forest fires are increasing, which may trigger multiple ecological crises. Forest fires can be scientifically prevented, and their risks can be mitigated through specific approaches, particularly by managing forest combustible materials. Common methods include mechanical clearance, prescribed burning, and the establishment of biological firebreak belts, along with the application of grazing to regulate forest fuels. This paper presents a review of studies on grazing and fire risk, both domestically and internationally. Research indicates that livestock grazing has complex effects on forest fire risk: appropriate grazing can manage fuels and modify ecosystem structure to reduce fire hazards—for instance, by decreasing the accumulation of surface flammable materials and promoting the regeneration of fire-resistant tree species. Conversely, overgrazing may disrupt ecological balance and increase fire risk, such as by exacerbating soil erosion and encouraging the invasion of flammable weed species. Case studies from different ecological regions worldwide demonstrate varied effects of grazing on fire prevention, though research in this area exhibits geographical disparities. Adaptive management should integrate targeted grazing, prescribed burning, and mechanical treatments in a synergistic manner. Future efforts should prioritize cross-scale studies, investigate the mechanisms of woody fuel modulation, and refine fire ecology models to enhance the precision and global applicability of grazing-based fire management.

Keywords: livestock grazing; forest fire risk; vegetation structure; fuel management; adaptive management strategies

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1. Introduction

Forest fires, as a critical natural disturbance factor in ecosystems, are exhibiting a global trend of increasing frequency, intensity, and destructiveness under the combined drivers of climate change and human activities such as land-use change and fire suppression legacies [1]. According to the global fire monitoring system, the average annual burned area over the past decade has risen significantly, resulting not only in substantial direct economic losses but also triggering cascading ecological crises such as biodiversity loss, decline in carbon sink function, and severe deterioration of air quality with far-reaching public health impacts [2]. In response to this escalating threat, grazing activities—through

livestock foraging, trampling, and other behaviors—modify vegetation density, height, and species composition, thereby enabling dynamic management of forest fuels [3].

The regulation of forest fire risk through grazing is essentially the result of interactions between livestock behavior and multifaceted ecosystem responses. Numerous studies consistently demonstrate that herbivory reduces fuel load and vegetation height through animal trampling and biomass consumption, thereby playing a crucial role in mitigating fire intensity and severity [4,5]. In southeastern Romania, for instance, large herbivores effectively consume fine surface fuels (e.g., grasses), directly reducing fire fuel availability and spread potential [6]. Furthermore, the vegetation patchiness created by grazing activities functions as a natural firebreak, decreasing the likelihood of fire ignition and propagation [7]. Selective foraging by livestock—such as goats consuming significant quantities of flammable shrub biomass—directly reduces surface fuel loads and disrupts spatial fuel continuity [8]. Grazing also indirectly enhances ecosystem fire resilience by altering plant community composition (e.g., promoting fire-resistant grass species) [9], modulating microclimatic conditions (e.g., increasing surface humidity) [10], and improving soil permeability [11]. However, when vegetation cover is excessively reduced or bare soil exposure becomes widespread, overgrazing can induce soil erosion [12] and facilitate the invasion of highly flammable annual weeds (e.g., *Vulpia megalura*) [13], ultimately expanding high fire-risk areas.

Current empirical research exhibits notable regional imbalances and methodological limitations. While substantial case studies have been accumulated in regions such as the Mediterranean basin, southern Spain, and southeastern France [14,15], many other critical fire-prone areas remain severely understudied. Methodologically, although research approaches have evolved from early plot-scale observations to cross-scale technologies such as LiDAR-based quantification of vegetation heterogeneity [16] and simulation of grazing-fire interactions [17–19], significant knowledge gaps persist regarding the regulatory mechanisms governing woody fuels (e.g., snags) and the long-term impacts of grazing on forest succession trajectories. This review synthesizes the multifaceted effects of livestock grazing on forest fire risk, critically examining its dualistic and context-dependent nature. While appropriate grazing practices can play a positive role in fuel reduction and the optimization of plant community structure, effectively mitigating forest fire hazards, overgrazing disrupts ecological balance and can substantially increase fire risk. Although case studies from various global ecosystems demonstrate the diverse potential of grazing in fire management, the existing research body is geographically skewed, raising questions about the universal applicability of current findings.

The escalating global wildfire crisis is not merely a consequence of increasing temperatures but a synergistic product of complex socio-ecological transitions. These include the expansion of the Wildland–Urban Interface (WUI), changes in land-use patterns such as rural abandonment in Mediterranean regions leading to fuel accumulation, and the spread of invasive flammable species. Traditional fire suppression methods, while crucial, are often reactive, cost-intensive, and can have negative ecological impacts. In this context, there is a growing imperative to develop proactive, cost-effective, and ecologically integrated fuel management strategies. Livestock grazing emerges as a promising 'nature-based solution' within this paradigm, not only due to its direct fuel reduction capacity but also because it offers a multifunctional approach. This approach aligns wildfire risk mitigation with broader sustainability goals, such as supporting rural economies, maintaining cultural pastoral landscapes, and promoting biodiversity in fire-adapted ecosystems. Consequently, this review aims to move beyond a simple assessment of grazing's effectiveness. It seeks to critically evaluate the conditions under which grazing transitions from a fire mitigation

tool to a potential ecosystem stressor, and to synthesize the principles for its sustainable integration into adaptive, climate-resilient forest management frameworks.

2. Multidimensional Impacts of Grazing on Forest Fire Regimes

Livestock grazing exerts multifaceted influences on ecosystems through behaviors such as foraging and trampling. These effects collectively impact various factors associated with forest fire risk. The underlying mechanisms can be categorized into direct mechanisms—primarily through fuel load management—that reduce fire risk, and indirect mechanisms involving ecosystem modification that also contribute to risk mitigation. However, under certain conditions, a risk reversal mechanism may occur, whereby grazing instead increases fire susceptibility.

2.1. Fuel Load Management

The most direct impact of grazing on forest fire risk lies in its effects on surface fuel load, structure, and spatial continuity. Livestock exhibit selective foraging behavior; for instance, goats and sheep preferentially consume herbaceous plants, young shoots of shrubs, and surface litter [20]. This sustained foraging activity significantly reduces the accumulation of flammable materials, thereby directly diminishing the available fuel following ignition. Studies have shown that in Andalusia, Spain, integrating livestock grazing as a silvopastoral practice effectively reduces surface fuel loads, offering a feasible, ecological, and economical strategy for clearing flammable vegetation within firebreaks [21].

Moreover, the movement and foraging behavior of livestock do not merely reduce fuel volume but actively engineer spatial patterns across the landscape. This grazing-induced heterogeneity naturally forms a network of "fuel breaks" or discontinuities [15]. These gaps are not random but often follow terrain or vegetation gradients influenced by animal preference and movement patterns. They effectively disrupt the horizontal continuity of combustible materials, thereby blocking the propagation pathways of surface fires and suppressing both the eventual burn area and the speed of fire progression [21]. This process of establishing ecological firebreaks through grazing transcends simple biomass removal; it represents a dynamic, self-adjusting form of biological fire management engineering. Its cost-effectiveness stems from leveraging natural animal behavior to achieve a management outcome that would otherwise require significant mechanical or chemical intervention.

2.2. Ecosystem Modification

Grazing activities initiate a cascade of ecological modifications that collectively alter fire regimes. The selective consumption of highly palatable and often flammable grass and shrub species reshapes plant community composition. This process creates a competitive advantage for fire-resistant plants (e.g., those with high moisture content, waxy leaves, or secondary compounds) or less preferred species [22]. Over time, this shift inherently reduces the overall combustibility of the ecosystem, steering the trajectory of plant succession towards communities with higher innate fire resilience [23]. This compositional change is intimately linked to structural and micro-environmental alterations. The reduction in herbaceous cover and litter layer, while increasing ground-level light, also diminishes total transpirational water loss and reduces competition for soil moisture. The resultant microclimate near the soil surface tends toward higher relative humidity and more stable soil moisture levels, creating conditions that increase the energy required for fuel ignition and slow pre-heating [24,25]. Furthermore, the return of livestock manure enhances the physicochemical properties of soil, increasing organic matter content and promoting soil aggregation. These improvements in soil structure bolster water retention capacity [26,27]. Consequently, vegetation in moderately grazed areas may maintain higher moisture con-

tent for longer during dry periods, effectively extending the annual "fire-safe period" and narrowing the window of high ignition risk.

2.3. Overgrazing

Based on the scientific principles of grazing management, grazing intensity can be characterized through three key parameters: intensity, the percentage of plant material removed; frequency, how often an area is grazed within a growing season; and timing, the plant growth stage during grazing. Moderate grazing is typically characterized by the removal of less than 50% of photosynthetic leaf area during periods of adequate soil moisture, allowing sufficient regrowth time for plants to replenish energy reserves. In contrast, overgrazing involves the repeated removal of more than 50% of leaf area, particularly during drought conditions or critical growth stages (e.g., boot stage), which depletes root reserves and hinders plant recovery [28].

When grazing pressure exceeds these ecological thresholds, its effects on fire risk are not universally beneficial but can reverse. Overgrazing leads to extreme reduction in vegetation height and density, which not only removes herbaceous fuels but also eliminates the protective plant cover crucial for water retention, thereby exacerbating soil erosion and degradation [29,30]. In arid and semi-arid regions, overgrazing creates ideal open ground for the invasion and establishment of flammable annual weeds such as *Bromus tectorum* [31]. These invasive grasses have short life cycles, rapidly senesce during dry periods, and form continuous, homogeneous fuel beds of high flammability, significantly increasing the risk of rapidly spreading, high-intensity wildfires [32].

3. Regional Case Studies: Divergent Grazing Outcomes in Global Fire-Prone Ecosystems

3.1. Comparative Analysis of Regional Cases

Different ecosystems exhibit highly specific responses to grazing due to their distinct vegetation structures, climatic conditions, and dominant fuel types. An integrated summary of these responses, including core characteristics, key empirical findings, and common research approaches for each major fire-prone ecosystem, is presented in Figure 1.

Research on Mediterranean shrublands is particularly well-established, providing a robust evidence base for grazing as a fire management tool. These regions are characterized by hot, dry summers and vegetation dominated by highly flammable sclerophyllous shrubs. Within this context, grazing is widely regarded as an effective bio-management tool for reducing flammable biomass and mitigating wildfire risk [33,34]. Studies on fire management in Mediterranean ecosystems have employed diverse methodologies: some researchers have conducted controlled field experiments using enclosures to compare fire prevention outcomes between grazed and ungrazed areas [35,36]; others have utilized spatial analysis and statistical models to examine correlations between historical fire data and livestock distribution patterns [37,38]; yet others have applied predictive modeling to simulate changes in fire behavior following grazing interventions [39]. A body of evidence indicates that grazing—particularly by goats and mixed herds—significantly reduces shrub biomass and disrupts fuel continuity [40–42]. A critical insight from this region is the role of land abandonment as a driver of fuel accumulation and increased fire frequency [43,44], highlighting grazing not just as an intervention but as a component of sustainable land-use that maintains low-fuel landscapes. However, the use of grazing as a fire management tool also entails risks: improperly executed prescribed burns by pastoralists may inadvertently serve as ignition sources [45,46]. Empirical studies demonstrate that effectively managed grazing can substantially mitigate fire behavior, reducing fire intensity and decreasing burned area by 25.9% to 60.9% [47].

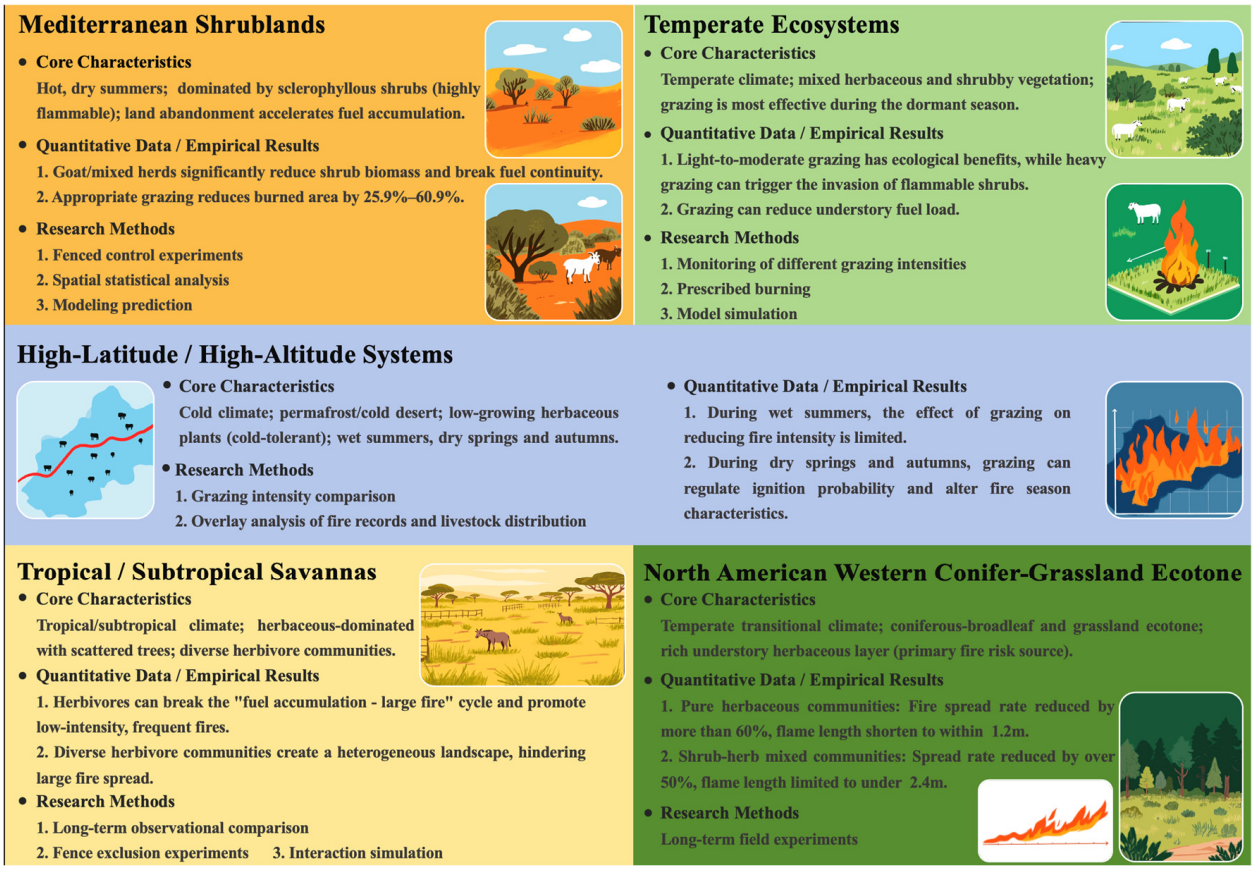


Figure 1. The Influence of Grazing on Fire Management in Different Ecosystems.

In temperate ecosystems, diverse methodological approaches have been employed to investigate grazing effects, including the establishment of permanent plots under varying grazing intensities to monitor vegetation dynamics [48], the implementation of prescribed burning or simulation-based modeling studies [49,50], and the analysis of correlations between livestock husbandry changes and historical wildfire records. Studies indicate that light to moderate grazing can yield ecological benefits, whereas heavy grazing may facilitate the invasion of flammable shrub species [48]. Furthermore, grazing has been shown to reduce understory fuel loads, although careful management of stocking density is essential to achieve desired outcomes [51]. Notably, implementing moderate grazing during the vegetation dormant season has been demonstrated to be particularly effective for wildfire risk mitigation, as it reduces dead fuel load without excessively compromising the protective cover and regenerative capacity of perennial grasses [52]. A key consideration in temperate systems is balancing fuel reduction with the maintenance of sufficient vegetation structure to prevent soil erosion and suppress the establishment of invasive annuals, a balance that requires careful, context-specific stocking rate management.

In high-latitude and high-altitude systems, spatial overlay analysis of historical fire records and livestock distribution data reveals a distinct season-dependent relationship between grazing activities and fire occurrence [53,54]. Comparative assessments of fire regime metrics—such as fire frequency and burned area—across regions with different grazing intensities indicate that the effectiveness of grazing in reducing fire intensity may be more limited in these regions' more humid summers, but its role in managing fine fuels in drier spring or autumn periods, thereby altering fire seasonality or limiting fire starts, can be significant [55]. This underscores that in such systems, grazing's primary fire mitigation benefit may lie in modifying ignition potential and fire timing rather than uniformly suppressing peak-season fire behavior.

In tropical and subtropical savannas, numerous studies have been conducted on fire dynamics and ecological regulatory mechanisms. For instance, long-term observations and comparisons of fire regimes have been carried out in parks or ranches with different herbivore assemblages [56,57]. Fence-exclusion experiments have been implemented to assess the impact of large herbivore removal [5], while modeling approaches have been employed to simulate the interactions among herbivores, vegetation, and fire [48,58]. Research findings indicate that herbivory can disrupt the "fuel accumulation to large fire" cycle and promote frequent, low-intensity fires [59,60]. Furthermore, diverse herbivore communities can create heterogeneous landscapes that impede the spread of large wildfires [56,61].

In the coniferous forest-grassland ecotones of western North America, fire risk is primarily driven by abundant herbaceous understory fuels [5]. A long-term experiment in southeastern Arizona demonstrated that grazing significantly suppresses fire behavior: within pure grass communities, the rate of fire spread was reduced by more than 60%, and flame length was shortened to within 1.2 m (reaching the critical threshold for effective suppression); in mixed grass-shrub communities, the rate of spread decreased by over 50%, with flame length limited to under 2.4 m [62]. By altering vegetation structure, grazing effectively reduces both the rate of spread and intensity of fires, thereby substantially lowering the difficulty of fire suppression efforts.

3.2. Analysis of Geographical Biases in Current Research

Current studies on the relationship between grazing and fire risk exhibit significant geographical imbalances. The majority of empirical evidence originates from regions with long-standing pastoral traditions and well-established ecological research frameworks, particularly Mediterranean-climate zones (e.g., Spain, Greece, France) and temperate grassland-forest ecotones in North America. These areas face substantial wildfire risks exacerbated by land-use changes and invasive species and have increasingly incorporated targeted grazing into official wildfire management strategies.

By contrast, significant gaps exist in research on other critical fire-prone regions globally. In tropical rainforest areas, such as the Amazon Basin and Southeast Asia, lands subjected to slash-and-burn agriculture are often converted to pasture; however, the impact of livestock grazing on fire risk dynamics in secondary forests remains poorly understood due to a lack of systematic monitoring [5]. Similarly, regions of Central and North Asia suffer from a systemic absence of monitoring data and a weak foundation of historical research. This scarcity of region-specific studies not only heightens their vulnerability under climate change but also impedes the development of science-based guidance for implementing low-cost fire mitigation strategies—such as targeted grazing—tailored to local ecological conditions.

3.3. Synthetic Comparative Summary

The case studies presented reveal a clear pattern: the role of grazing in wildfire mitigation is not uniform but is fundamentally shaped by ecosystem-specific characteristics, particularly the dominant fuel types and climatic constraints. Table 1 synthesizes these divergent outcomes by highlighting the primary mechanisms and management implications in each major ecosystem.

In synthesis, grazing operates most directly and effectively in ecosystems where the primary wildfire threat stems from herbaceous or fine woody fuels that are palatable and accessible to livestock (e.g., Mediterranean shrublands, savannas, forest-grassland ecotones). Its role becomes more nuanced or seasonally constrained in systems where fuels are predominantly woody, less palatable, or where climatic humidity limits combustion potential. A universal theme is the non-linear relationship between grazing pressure and

fire risk, where moderate, well-timed grazing reduces hazard, but both undergrazing (fuel accumulation) and overgrazing (ecosystem degradation, weed invasion) can increase it.

Table 1. Comparative overview of grazing mechanisms and key implications across fire-prone ecosystems.

Ecosystem	Dominant Fuels/Context	Primary Role of Grazing in Mitigation	Key Management Insight
Mediterranean Shrublands	Dense, flammable sclerophyllous shrubs.	Direct reduction in shrub biomass and disruption of fuel continuity.	A key tool to counteract fuel accumulation from land abandonment; requires careful stocking rate management.
Temperate Ecosystems	Herbaceous understory; mixed grass-shrub layers.	Reduction in fine fuel loads, particularly when applied during the dormant season.	Effectiveness hinges on moderate, seasonally timed grazing to avoid soil erosion or invasive species promotion.
High-Latitude/ Altitude Systems	Fine fuels in dry seasons; humid summers.	Modifying ignition potential and fire seasonality by managing fine fuels in spring/autumn.	Benefit is seasonal; focus should be on pre-emptive fuel management in defined risk windows rather than peak summer.
Tropical/Subtropical Savannas	Continuous grassy fuels.	Disrupting the grass-fuel accumulation cycle and promoting landscape heterogeneity.	Integrating herbivory (wild or domestic) with fire regimes can sustain a mosaic that limits large wildfires.
Coniferous Forest-Grassland Ecotones	Abundant herbaceous understory fuels.	Direct consumption of grassy fuels, reducing rate of spread and intensity.	A highly effective strategy where cattle grazing can be targeted to specific high-risk interface zones.

4. Implementing Adaptive Grazing Strategies for Wildfire Management

4.1. Targeted Regulation Strategies

A one-size-fits-all grazing approach is inadequate to address the complex and dynamic patterns of forest fire risk. It is essential to match livestock species with the dominant fuel types present in different regions. The foraging behavior and physiological characteristics of animals determine their optimal habitats. Goats, being skilled climbers with a preference for low-fiber leaves and woody shrubs [63], are particularly suitable for managing brush and flammable understory shrub layers. Cattle, as non-selective grazers, efficiently and uniformly consume large areas of herbaceous fuels [64], making them well-suited for mitigating fire risk in forest-grassland ecotones and grassland regions. Sheep, which feed on both grasses and low shrubs and are highly agile [64], are more appropriate for grazing in shrub-grass transition zones or areas with fragmented topography.

Furthermore, stocking rate must be dynamically adjusted based on fire risk levels. Utilizing remote sensing, vegetation surveys, and fire behavior models, a "fire risk zoning map" can be developed to guide the flexible allocation of livestock. Adopting a fixed stocking rate may result in undergrazing in some areas and overgrazing in others. For example, designating extreme-risk zones as priority grazing areas while reducing stocking rates or implementing temporary grazing exclusion in low-risk regions can enhance management efficacy [65]. This approach, known as "targeted grazing", ensures that management actions are focused on the most critical fire prevention nodes, significantly improving resource utilization efficiency [66]. Implementing targeted grazing therefore requires a dynamic, data-driven approach that moves beyond static stocking rates. For example, a "prescriptive

grazing" plan can be developed by integrating near-real-time satellite imagery to identify areas of high fuel accumulation (e.g., using NDVI or specific fuel indices) with spatially explicit fire weather risk models (e.g., Fosberg Fire Weather Index). Livestock can then be strategically concentrated in these high-priority zones during periods of peak fire danger. Conversely, stocking density can be reduced or animals rotated out during ecologically sensitive periods, such as plant flowering, seeding, or wildlife nesting seasons, or in areas assessed as low risk. This dynamic allocation transforms grazing from a blanket treatment into a precision tool, responsive to both ecological conditions and fluctuating fire hazard.

4.2. Synergistic Integration of Multiple Measures

4.2.1. Prescribed Burning

The integration of grazing with prescribed fire can follow a strategic sequence: (1) Pre-fire grazing to reduce fine fuel loads and ensure burn controllability; (2) Prescribed fire conducted under optimal conditions to consume ladder fuels and stimulate nutrient cycling; (3) Post-fire grazing to manage the rapid growth of often-flammable early successional vegetation, thereby extending the treatment's effective lifespan. This "grazing-fire-grazing" cycle mimics natural disturbance regimes and can enhance habitat heterogeneity.

Short-term high-intensity grazing implemented prior to prescribed burning can effectively reduce fine surface fuel loads, thereby significantly decreasing fire intensity and variability during controlled burns. This enhances operational safety and controllability [67]. Meanwhile, prescribed burning promotes the decomposition of plant residues and accelerates nutrient cycling, leading to improved forage quality in the early grazing season. Grazing in recently burned areas not only provides high-quality forage for livestock but also helps control rapidly regenerating flammable grasses and weeds after the fire. This complementary approach extends the duration of fire prevention effectiveness [68,69].

4.2.2. Mechanical Fuel Reduction

Grazing activities face geographic constraints in areas such as steep slopes, rocky terrain, or ecologically sensitive zones (e.g., riparian corridors and habitats of endangered species). In these locations, mechanical fuel reduction serves as a critical complementary measure. By using methods such as mechanical mastication and shredding to pre-establish safety zones or fuel breaks [70], livestock can then be directed to graze in these treated areas under controlled conditions. This approach forms an integrated fire management model combining "mechanical pretreatment with biological regulation," enhancing both safety and ecological effectiveness.

5. Discussion

5.1. Adaptive Grazing as a Multifunctional Fire Management Tool

Appropriate grazing serves as an effective and multifaceted strategy for managing forest fuels. Through direct biomass consumption and trampling, livestock reduce surface fuel loads and disrupt horizontal fuel continuity, thereby impeding fire ignition and spread [71]. Beyond this immediate fuel-modifying effect, moderate grazing can positively reshape ecosystem structure and function [72]. By selectively suppressing flammable shrubs and weeds while promoting the regeneration of fire-resistant species, grazing contributes to a long-term reduction in overall landscape combustibility. Furthermore, grazing influences micro-environmental conditions; improved soil structure and enhanced vegetation cover can increase soil moisture retention [73,74] and reduce the accumulation of dry surface fuels, creating a less fire-prone environment.

However, the relationship between grazing and fire risk is inherently non-linear and context-dependent. Overgrazing can trigger a reversal of these benefits, leading to veg-

etation cover loss, soil erosion, and the invasion of highly flammable annual weeds [75], ultimately exacerbating fire hazard. This duality underscores that grazing is not a universally beneficial intervention but a management tool whose outcomes are dictated by intensity, timing, livestock species, and ecological context [76].

The synthesis of case studies from diverse ecosystems (summarized comparatively in Figure 1) demonstrates this variability. While grazing is widely recognized as a beneficial fire management tool in Mediterranean systems [77], its role in humid or high-latitude regions may be more subtle or seasonally constrained [78]. A persistent challenge is the pronounced geographical bias in existing research, with significant knowledge gaps in critical fire-prone regions like tropical rainforests and boreal zones. This imbalance limits the development of robust, globally applicable frameworks.

Consequently, moving beyond a one-size-fits-all approach is imperative. The synthesis of global case studies clearly indicates that successful fire mitigation through grazing depends on a carefully calibrated, adaptive strategy. This involves not only aligning livestock species with specific fuel types and dynamically adjusting stocking rates based on fire risk zoning but also strategically sequencing grazing with other interventions within a broader management framework [79]. For example, short-term intensive grazing prior to prescribed burning can enhance its safety and effectiveness by moderating fire behavior, while post-fire grazing can control the rapid regrowth of often-flammable early successional vegetation, thereby extending the treatment's effective lifespan. In topographically complex or ecologically sensitive areas where grazing access is limited, initial mechanical fuel reduction can create the necessary conditions to enable safe and effective subsequent grazing. This synergistic, "ecological engineering" approach leverages the complementary strengths of each method—biological regulation, controlled fire, and mechanical treatment—to establish a more resilient, cost-effective, and ecologically nuanced fire management system.

5.2. Research Gaps and Future Directions

To translate the potential of grazing into reliable, scalable fire mitigation practice, future research must address several interconnected knowledge gaps and methodological frontiers.

First, a deeper mechanistic understanding is required. While the effects of grazing on herbaceous fuels are relatively well-documented, its indirect regulation of woody fuels (e.g., standing dead trees, coarse woody debris) remains poorly understood. Research should investigate whether livestock activities, through trampling or microenvironment modification, can accelerate the decomposition of these high-energy-density fuels [80]. Furthermore, systematic studies are needed to quantify the differential impacts of livestock species (e.g., goats, cattle, sheep) and define ecologically sustainable stocking rate thresholds across diverse ecosystems, particularly in understudied regions [81]. A key focus must be identifying the tipping points where grazing pressure shifts from reducing to amplifying fire risk.

Second, advancing predictive capacity demands cross-scale and cross-disciplinary integration. Current fire behavior models inadequately represent the dynamic and multifactorial effects of grazing. Future work should develop coupled grazing-fire-vegetation models that dynamically integrate livestock foraging pressure, species-specific impacts, and associated feedbacks on plant community composition, fuel structure, and microclimate [82]. This requires blending field ecology with remote sensing (e.g., LiDAR for 3D fuel mapping), long-term landscape-scale monitoring, and computational simulation [83]. Additionally, integrating social science perspectives is crucial to understanding stakeholder perceptions, economic viability, and the governance models that facilitate or hinder the adoption of fire-smart grazing practices.

Finally, bridging science with policy and practice represents a critical translational frontier. Research must expand beyond biophysical outcomes to examine the socio-economic dimensions of grazing-based fire management [84]. This includes conducting cost–benefit analyses comparing grazing to alternative fuel treatments [85], designing incentive structures and insurance products for landowners, and exploring how traditional and local ecological knowledge can inform adaptive co-management frameworks [86]. Developing clear implementation guidelines—from risk-based zoning and livestock selection to monitoring protocols and adaptive feedback loops—will be essential for operationalizing targeted grazing [87]. By closing these research gaps, the scientific community can provide land managers and policymakers with the robust, context-sensitive evidence needed to harness grazing as a cornerstone of integrated, adaptive, and sustainable wildfire risk reduction strategies [88].

6. Conclusions

The relationship between grazing and forest fire risk is complex and context-dependent, governed by an interplay of ecological, climatic, and management factors. As this review illustrates, livestock grazing holds significant potential as a multifunctional, nature-based strategy for mitigating wildfire risk, primarily through direct fuel reduction and indirect ecosystem modification. However, this potential is tightly constrained by the risks of overgrazing, which can degrade ecosystems and inadvertently increase fire hazard. The divergent outcomes observed across global regions underscore that there are no universal prescriptions. Therefore, the future of grazing in fire management lies in embracing and implementing adaptive, precision frameworks. Success depends on moving beyond broad generalizations to develop locally tailored strategies that strategically combine targeted grazing with other fuel treatments. Prioritizing research to close critical geographical and mechanistic knowledge gaps, particularly in underrepresented ecosystems and regarding woody fuel dynamics, is essential to refine predictive models and provide robust, evidence-based guidance. By doing so, land managers and policymakers can better harness grazing not merely as an agricultural or conservation practice but as a vital, flexible component of integrated, resilient, and sustainable landscape management.

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